Aamir Malik

Machine Learning and Computational Materials Scientist, PhD (KAIST)
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Summary

Materials scientist (PhD, KAIST, 2025) with 5+ years combining **machine learning** (graph neural networks, Bayesian optimization, active learning, generative models, computer vision) and **first-principles simulation** (DFT/VASP on national HPC) to design and screen materials. Built an automated Python workflow for high-throughput VASP calculations on the KISTI Nurion supercomputer, and maintains about 30 public, reproducible ML, simulation and LLM-agent repositories benchmarked against tuned baselines. Co-first author on a peer-reviewed journal article; 11 international conference presentations. Background in operations research (MS, KAIST) and university technology commercialization. Seeking AI/ML scientist, applied scientist and computational materials roles. Pakistani national based in Seoul; **available to relocate to the United States** and able to begin employment as soon as a visa (J-1 or H-1B) is approved once an offer is in place.

Skills

- **Machine learning:** Python, PyTorch, TensorFlow, Keras, scikit-learn, NumPy, pandas, SciPy; deep learning, graph neural networks, Gaussian processes, Bayesian optimization, active learning, transfer learning, Bayesian neural networks, generative models (VAE, GAN), CNN and U-Net computer vision, positive-unlabeled learning.
- **Simulation and computational materials:** VASP (DFT+U, spin-polarized), Gaussian, CP-PAW, computational hydrogen electrode, Bader charge analysis, COHP bonding analysis (LOBSTER), pymatgen, ASE, machine-learned interatomic potentials (CHGNet, MACE), LAMMPS, ANSYS (FEM).
- **Large language model (LLM) agent systems:** bounded tool-using agents (typed tool schemas, capability contracts, budgets, checkpoint and resume, trace replay), agent evaluation harnesses with baselines and held-out tasks, prompt-injection and fault-injection testing.
- **Agentic AI tools:** Claude Code, OpenAI Codex, Cursor, GitHub Copilot, Gemini CLI, Kimi, OpenClaw; Model Context Protocol (MCP) tool integration; agentic coding and multi-agent workflows for tested, documented research software.
- **Data and engineering:** SQL, R, MATLAB, Bash, Linux, PBS job scheduling, HPC (KISTI Nurion), Git and GitHub, unit testing and CI, reproducible workflows.
- **Optimization and analytics:** CPLEX, Xpress MP, NSGA-II, decision analysis and risk management (Palisade DecisionTools), capacity planning and scheduling (AutoSched AP); materials characterization (TEM, SEM, XRD; KAIST Analysis Center self-user, 2022 – 2025).

Experience

Postdoctoral Researcher, Konkuk University, Seoul, Republic of Korea **Mar. 2026 – Present**

- Conduct DFT (VASP, DFT+U) studies of oxygen evolution reaction electrocatalysts for alkaline and seawater electrolysis within the computational hydrogen electrode framework, including surface stability, dopant site preference and reaction pathways.
- Built an automated Python workflow for high-throughput VASP calculations: input validation, structure generation (pymatgen, ASE), PBS job submission, convergence classification with corrective reruns, and a single traceable results ledger.
- Run large-scale calculations on the KISTI Nurion national supercomputer and laboratory PBS clusters.

Doctoral Researcher, KAIST, Daejeon, Republic of Korea **Mar. 2018 – Aug. 2025**
Nuclear Fuel Materials Laboratory (2020 – 2025); Optimization and Statistical Inference Laboratory (2018 – 2020)

- Developed graph neural network, Bayesian neural network and transfer-learning models for property prediction and synthesizability classification of TiZrNb-based high-entropy alloys, including a physics-inspired GNN architecture and advanced input-feature engineering.
- Implemented Bayesian optimization with thermodynamics-based active learning for multi-objective alloy design, closing the loop between model predictions and experimental synthesis; data-driven Cu alloy design for strength and electrical conductivity with experimental iteration.
- Applied generative models (GAN, VAE, AAE) and positive-unlabeled learning for inverse design and synthesizability prediction; NSGA-II for multi-objective optimization of strength and hardness.
- Built AI models for tire compound property prediction and inverse design of tire recipes in an industry collaboration project (2018 – 2020).

Deputy Director, Office of Research, Innovation, and Commercialization, GIKI **Sep. 2017 – Feb. 2018**

- Fostered public-private partnerships, aligned academic research with corporate priorities, and accelerated technology transfer and commercialization of university research.

Master's Researcher, Complex Stochastic and Service System Design Lab, KAIST **Mar. 2015 – Feb. 2017**

- Semiconductor fab system design and optimization; modular system design and optimization for an oil refinery.

Earlier: Materials Engineer, Department of Materials Science and Engineering, GIKI (Aug. 2013 – Aug. 2014; supervised laboratory courses for up to 100 students, trained students in Pro/ENGINEER, Creo and ANSYS); Production Intern, MOL Pakistan Oil and Gas (Jul. – Aug. 2012).

Selected Projects

- **Bounded LLM agent systems (2026):** five Python repositories on one shared agent core (typed tools with allowlists and budgets, checkpoint and resume, trace replay, no agent frameworks): an evidence agent with hash-verified citations, a benchmark harness for tool-using agents under capability contracts with fault and prompt injection, a recovery agent for failed scientific computations with approval-gated repair and rollback, a budgeted experiment-campaign agent, and a tabular ML experiment-planning agent; each evaluated against non-LLM and single-call baselines on held-out tasks. github.com/aamirmalik-dr/agent-contract-bench
- **Machine learning for electron microscopy (2026):** seven reproducible projects on simulated data with exact ground truth, benchmarked against tuned classical baselines (atom detection, denoising, defect segmentation, diffraction classification, EELS unmixing, 4D-STEM mapping, active learning); Gaussian-process active learning reached target map accuracy with about 21 percent fewer measurements than a space-filling scan. github.com/aamirmalik-dr/active-learning-microscopy
- **Machine-learned interatomic potentials for TiZrNb (2026):** from-scratch Behler-Parrinello and SchNet-style potentials, query-by-committee active learning (2.2x label saving on a scarce pool, random selection winning on a mixed pool, both reported), and a benchmark of a compact distilled potential against CHGNet and MACE as ASE calculators. github.com/aamirmalik-dr/alloy-mlip-bench

Education

- **PhD, Materials Science and Engineering**, KAIST, Daejeon; conferred Aug. 15, 2025 Feb. 2018 – Aug. 2025
- **MS, Industrial and Systems Engineering**, KAIST, Daejeon Mar. 2015 – Feb. 2017
- **BS, Materials Science and Engineering**, Ghulam Ishaq Khan Institute (GIKI), Topi Aug. 2009 – Jun. 2013
- **MITx MicroMasters** (Statistics and Data Science; Data, Economics, and Design of Policy; Finance), online Sep. 2023 – Dec. 2027

Professional Training

- **Sep. 2026:** Hands-on Course on Density-Functional Calculations, International CP-PAW Autumn School, TU Clausthal at the University of Göttingen, Germany (two weeks; team project on hydrogen diffusion in Pd released as a tested repository).
- **Jun. 2026:** 4th Summer School on Machine Learning for Electron Microscopy (MLSTEM 2026), University of Tennessee, Knoxville, USA (one week; followed by seven open-source microscopy ML projects).

Selected Publications

1. Muhammad Akmal*, **Aamir Malik***, Wonjong Jeong, Ho Jin Ryu, "Incorporating microstructural and mechanical heterogeneity to Ti-Zr-Nb alloy by partial high-energy ball milling", *Materials Chemistry and Physics*, Vol. 315, 129037, 2024 (*equal contribution, shared first authorship).
2. Muhammad Akmal, Avinash Chavan, **Aamir Malik**, Dinesh Somaya, Hyun Woo Seong, Jennifer Shin, Ho Jin Ryu, "Gradient-controlled interfaces in medium/high entropy alloy composites for enhanced mechanical and biocompatibility performance", *Applied Surface Science Advances*, Vol. 30, 100868, 2025.

Three first-author manuscripts on GNN-based inverse design and synthesizability prediction of high-entropy alloys to be submitted; 11 international and 8 domestic conference presentations, including MRS Fall Meeting 2024 (Boston), PacRim16 2025 (Vancouver) and CALPHAD 2025 (Busan).

Honors and Awards

KAIST scholarships for master's (2015) and doctoral (2018) studies; Dean's Honour Roll, GIKI (6 of 8 semesters).

Languages

English (professional), Urdu (native), Hindko (native), Korean (elementary).